



United Nations

**EKAP Digital Government Intelligence Platform
Architecture Plan, Database Schema, and Functional Specifications**

May 2026

Phase 3 - Technical Design Package

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EKAP Digital Government Intelligence Platform – Phase 3: Technical Design Package

Prepared for the **United Nations Department of Economic and Social Affairs
Division for Public Institutions and Digital Government (DPIDG), UN DESA, and
Digital Government
Digital Government Branch (DGB)**

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DISCLAIMERS

Purpose and Status

This Technical Design Package is an analytical and design document prepared under the consultancy "Feasibility Study and Prototype Development for the UN Digital Government Intelligence Platform (EKAP)". It constitutes Deliverable 3 as defined in the Terms of Reference (TOR OP5.1 EKAP), with a submission deadline of 19 December 2025. It provides the architecture plan, database schema, and functional specifications for EKAP, aligned with UN OICT standards.

Primacy of the ToR

The Terms of Reference (TOR OP5.1 EKAP) remain the authoritative reference for the scope of work, deliverables, and schedule. Where any apparent inconsistency exists between this Technical Design Package and the ToR, the ToR takes precedence.

Relationship to Prior Deliverables

This document builds directly on the Inception Report (Deliverable 1) and the Feasibility Study Report (Deliverable 2). It implements the recommended hybrid architecture and functional specifications derived from those prior deliverables and confirmed through stakeholder engagements, including the E-Gov Survey System Implementation Meeting of 14 March.

Decision-making and Procurement

The specifications and designs presented here are inputs to UN DESA and UN OICT decision-making. Final choices regarding implementation approach, procurement, and hosting require formal UN authorization and procurement processes. This document does not pre-empt those processes.

Confidentiality

This Technical Design Package and any associated drafts and working materials are internal to DPIDG, DGB, UN OICT, and authorized partners. Public dissemination requires formal clearance in line with UN DESA publication policies.

**ACRONYMS AND ABBREVIATIONS**

Abbreviation	Definition
AI	Artificial Intelligence
AOSQ	Agent-assisted Online Services Questionnaire layer
API	Application Programming Interface
ARB	Architecture Review Board (UN OICT)
BI	Business Intelligence
COTS	Commercial Off-The-Shelf
DGB	Digital Government Branch
DPIDG	Division for Public Institutions and Digital Government
EGDI	E-Government Development Index
EGovKB	E-Government Knowledge Base
EGOVSURVEY	UN E-Government Survey Platform (current system)
EKAP	E-Government Knowledge and Assessment Platform (next-generation)
EPI	E-Participation Index
ER	Entity-Relationship
HCI	Human Capital Index
IAM	Identity and Access Management
IF	Infrastructure Index (OSI sub-component)
IIS	Internet Information Services (Microsoft web server)
ITU	International Telecommunication Union
LDC	Least Developed Countries
LGQ	Local Government Questionnaire
LLDC	Landlocked Developing Countries
LOSI	Local Online Services Index
LOSI-C	LOSI for Cities
LOSI-T	LOSI for Towns and Semi-Dense Areas
LOSI-V	LOSI for Villages and Rural Areas
MSQ	Member State Questionnaire
OICT	Office of Information and Communications Technology (United Nations)
OSI	Online Services Index
OSQ	Online Services Questionnaire
RBAC	Role-Based Access Control
SIDS	Small Island Developing States



SP	Service Provision (OSI sub-index)
TEC	Technology (OSI sub-component)
TII	Telecommunication Infrastructure Index
ToR	Terms of Reference
UN DESA	United Nations Department of Economic and Social Affairs
UNDP	United Nations Development Programme
UNESCO	United Nations Educational, Scientific and Cultural Organization
VPN	Virtual Private Network



1. Executive Summary

EKAP (E-Government Knowledge and Assessment Platform) is a next-generation, cloud-based, AI-enabled digital government intelligence platform commissioned by the United Nations Department of Economic and Social Affairs (UN DESA) through the Division for Public Institutions and Digital Government (DPIDG) and the Digital Government Branch (DGB). EKAP consolidates the existing EGOVSURVEY assessment platform and the EGovKB knowledge base into a single governed environment, extending capabilities to meet the requirements established in the Terms of Reference (TOR OP5.1 EKAP).

This Technical Design Package constitutes Deliverable 3 and provides the architecture plan, database schema, and functional specifications aligned with UN OICT standards. It builds on the Feasibility Study (Deliverable 2) and the Inception Report (Deliverable 1), implementing the recommended hybrid approach.

Strategic Objectives

- Unify survey operations, index management, and knowledge outputs at national and local levels within a single platform.
- Modernize the assessment workflow covering OSQ, MSQ, and LGQ instruments, with integrated OSI and LOSI calculation.
- Introduce a formal Government Actor role enabling authorized officials to update MSQ and LGQ data under UN review.
- Enable structured assessor management for approximately 400 internal and external assessors.
- Automate index measurement for OSI, LOSI, and EGDI, replacing manual spreadsheet-based calculations.
- Align all components with UN OICT approved standards, security policies, and cloud infrastructure.
- Integrate AI-assisted features (AOSQ) under strict governance and human oversight.

Architecture Approach

The preferred direction, as recommended in the Feasibility Study, is a hybrid solution comprising:

- A custom EKAP core application implementing the OSI/LOSI toolkit, question lifecycle management, assessor and Government Actor roles, and AI/AOSQ integration hooks.
- COTS and platform services selected from UN OICT approved products for BI, dashboards, analytics, and content delivery.
- The platform will feature cloud hosting on Microsoft Azure or AWS using OICT-approved services, ensuring compatibility with the Microsoft stack (IIS, SQL Server, .NET)



Key Design Decisions

Decision	Position	Source
Architecture model	Hybrid: custom EKAP core plus COTS BI and analytics services	Feasibility Study, Section 9.5
Hosting platform	Azure or AWS with OICT-approved services; Microsoft stack compatible	Feasibility Study, Section 8.5
Core data store	SQL Server (Azure SQL Database or IaaS equivalent)	DGB Infrastructure; OICT standards
Identity management	UNHQ Active Directory (NOT Azure AD); VPN-based admin access	DGB Infrastructure document
BI and dashboards	Power BI (Power Platform ARB 2025-112) or Microsoft Fabric (ARB 2023-393)	Current_standards.pdf
AI services	Azure OpenAI Service (ARB 2023-158) in limited, governed AOSQ scenarios	Current_standards.pdf
Baseline schema	EGOV SURVEY EKBSRVY tables extended with new EKAP entities	DGB Infrastructure document
Index computation	Internal OSI/LOSI toolkit replacing manual spreadsheet processes	Feasibility Study, Section 7

2. Introduction

2.1 Purpose of This Document

This Technical Design Package responds to the ToR requirement for "Architecture plan, database schema, and functional specifications aligned with UN OICT standards" (Deliverable 3, deadline 19 December 2025). It provides sufficient technical detail to guide subsequent prototype development, implementation tendering, and the eventual Implementation ToR (Deliverable 6).

2.2 Relationship to Prior Deliverables

The Inception Report established the consultancy scope, delivery schedule, and seven central business requirements confirmed in the E-Gov Survey System Implementation Meeting of 14 March. The Feasibility Study analyzed the as-is environment, evaluated architecture options, and recommended the hybrid approach. This Technical Design Package translates those findings into actionable specifications.

2.3 Scope and Boundaries

This document covers the following within the ToR scope:

- As-is architecture of EGOVSURVEY and EGovKB.



- Target logical and data architecture for EKAP (hybrid approach).
- Database schema: EGOVSURVEY baseline entities and EKAP schema extensions.
- Functional specifications for all major EKAP modules and workflows.
- Integration architecture for external data sources and downstream consumers.
- Security, RBAC, and UN OICT compliance requirements.
- Non-functional requirements.
- Data migration approach.
- Implementation roadmap aligned with the ToR schedule.

2.4 Document Authority and References

- TOR OP5.1 EKAP - Terms of Reference (primary authority)
- EKAP Inception Report 2.4 (Deliverable 1)
- EKAP Feasibility Study Report (Deliverable 2)
- Concept Paper 4.44 - EKAP strategic and methodological baseline
- Existing DGB Applications and Infrastructure (system inventory)
- UN OICT Software Registry (current_standards.pdf)

3. As-Is Architecture

This section documents the current technical state of EGOVSURVEY and EGovKB as the baseline for EKAP development. All information is sourced from the Existing DGB Applications and Infrastructure document.

3.1 Current System Inventory

System	Function	Technology Stack	Servers	Status
EGOVSURVEY	Survey assessment platform	DNN 9.6.1, ASP.NET WebForms, WebAPI	NYVM2500 (IIS), NYVM2501 (SQL Server)	Active
EGovKB	E-Government Knowledge Base	DNN 9.6.1, ASP.NET WebForms, WebAPI	NYVM2500 (IIS), NYVM2501 (SQL Server)	Active

3.2 Infrastructure Details

- **NYVM2500 (Web Server):** Running Internet Information Services (IIS).
- **NYVM2501 (Database Server):** Running Microsoft SQL Server; database: SurveyEgov.
- **Identity and Access:** Microsoft Active Directory (UNHQ domain); NOT Azure AD. UNHQ VPN required for all administrative access.
- **Gateway:** Citrix NetScaler gateway mediates all external web access.



3.3 Technology Stack

- DNN (DotNetNuke) 9.6.1 content management framework
- ASP.NET WebForms for survey and data-entry interfaces
- ASP.NET WebAPI for programmatic access
- Internet Information Services (IIS) on Windows Server
- Microsoft SQL Server for all relational data storage

3.4 Current Workflow Summary

The current end-to-end workflow, as documented in the Feasibility Study (Section 5.2):

- **Data collection:** OSQ, MSQ, and LGQ data collected through EGOVSURVEY interfaces from assessors and focal points.
- **Analysis:** Responses cleaned and validated using external tools including spreadsheets and scripts.
- **Index calculation:** OSI, LOSI, and related indices computed through semi-manual procedures, often outside EGOVSURVEY.
- **Publication:** Results exported from EGOVSURVEY, transformed, and loaded into EGovKB or other formats.
- **Visualization:** Dashboards and visualizations generated within EGovKB using relatively static templates.

3.5 Known Limitations

Functional fragmentation:

- EGOVSURVEY manages data collection; EGovKB presents results. Index calculation occurs outside both systems.
- No single orchestrated workflow from data capture to publication.

Technical constraints:

- Legacy DNN-based stacks and manual deployments.
- Limited cloud-native capabilities and partial alignment with modern IAM and security patterns.

Operational gaps:

- No comprehensive assessor registry, performance model, or training lifecycle inside the platform.
- No systematic "History of a Question" view spanning survey cycles.
- Limited automation for OSI/LOSI measurement and EGOVSURVEY-EGovKB integration.



4. Target Architecture

4.1 Architecture Approach: Hybrid Model

As recommended in the Feasibility Study (Section 9.5), EKAP follows a hybrid architecture. The hybrid scenario is preferred because:

- It allows EKAP to encode EGDI-specific index logic and question history as first-class design elements.
- It leverages COTS strengths for BI, dashboards, and generic workflows.
- It aligns with Microsoft stack and cloud options endorsed by UN OICT.
- It reduces long-term technical risk compared with a pure COTS or purely custom approach.

4.2 Design Principles

- Alignment with UN OICT standards for cloud, IAM, security, and data protection.
- Compatibility with the Microsoft stack (IIS, SQL Server, .NET), Azure, and AWS.
- Modular architecture separating survey operations, OSI/LOSI toolkit, knowledge base, analytics, and AI services.
- Data-centric design preserving EGOVSURVEY history and enabling longitudinal analysis.
- API-first integration approach for both internal and external consumers.
- Cloud-first hosting; separate development, testing, staging, and production environments required.
- Business ownership: DPIDG. Technical ownership: UN OICT.

4.3 Logical Architecture Layers

Layer 1 - User Experience Layer

- Assessor interfaces: OSQ submission, evidence upload, review workflows.
- Government Actor portal: MSQ and LGQ data update, submission tracking, approval status.
- UN staff interfaces: survey configuration, index governance, assessor management, admin dashboards.
- Public interfaces: EGDI/LOSI dashboards, country profiles, knowledge base via EGovKB.

Layer 2 - Application Layer

- Survey configuration module: cycle management, questionnaire builder, repopulation settings.
- OSQ/MSQ/LGQ workflow module: data entry, validation, and review.
- Assessor management module: recruitment, profiling, training, assignment, and performance.
- Government Actor module: portal, submission routing, approval chain.



- OSI/LOSI toolkit module: indicator definitions, question-indicator mappings, computation engine.
- Index governance module: configuration, automated calculation, data approval workflows.
- Knowledge management module: EGovKB integration, content APIs, toolkit linkages.
- AI/AOSQ services module: pre-assessment, companion agent, quality assurance (governed scenarios only).
- Reporting and analytics module: dashboards, exports, data feeds.

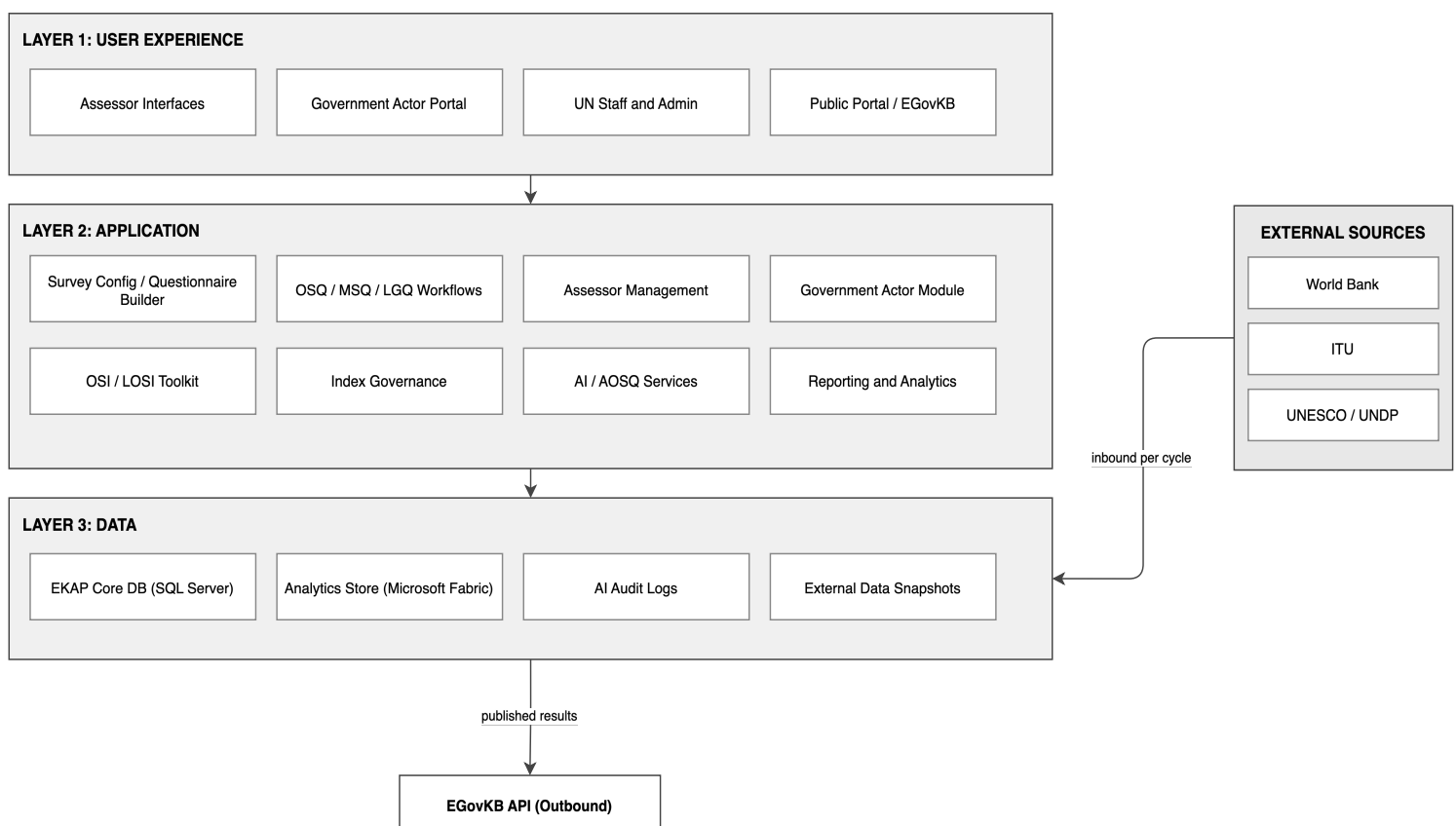
Layer 3 - Data Layer

- Unified relational database for core EKAP transactional data (SQL Server).
- Analytical data stores for BI and reporting (Microsoft Fabric or Azure Synapse, OICT-approved).
- AI interaction and audit logs for AOSQ transparency and traceability.
- External data snapshots for per-cycle reproducibility of EGDI and LOSI results.

Layer 4 - Integration Layer

- Inbound connectors: World Bank, ITU, UNESCO, UNDP, and other trusted external sources.
- Outbound APIs: EGovKB and internal UN reporting systems.
- Secure gateway and network segmentation consistent with UN security policies.
- Monitoring and logging aligned with OICT operational standards.

Figure 1: EKAP Logical Architecture





4.4 Hosting and Cloud Compatibility

- EKAP core hosted on Azure or AWS using OICT-approved services, leveraging the Microsoft stack.
- Azure SQL Database (SQL Server-compatible) as the primary relational data store.
- Cloudflare CDN (ARB 2024-134) for content delivery and perimeter protection.
- Google Web Analytics (ARB 2025-195) for public portal analytics where permitted.
- Grafana (ARB 2023-030) for operational monitoring and system metrics visualization.
- Separate environments required: development, testing, staging, and production (Concept Paper 4.44).

4.5 OICT-Approved Products in Scope

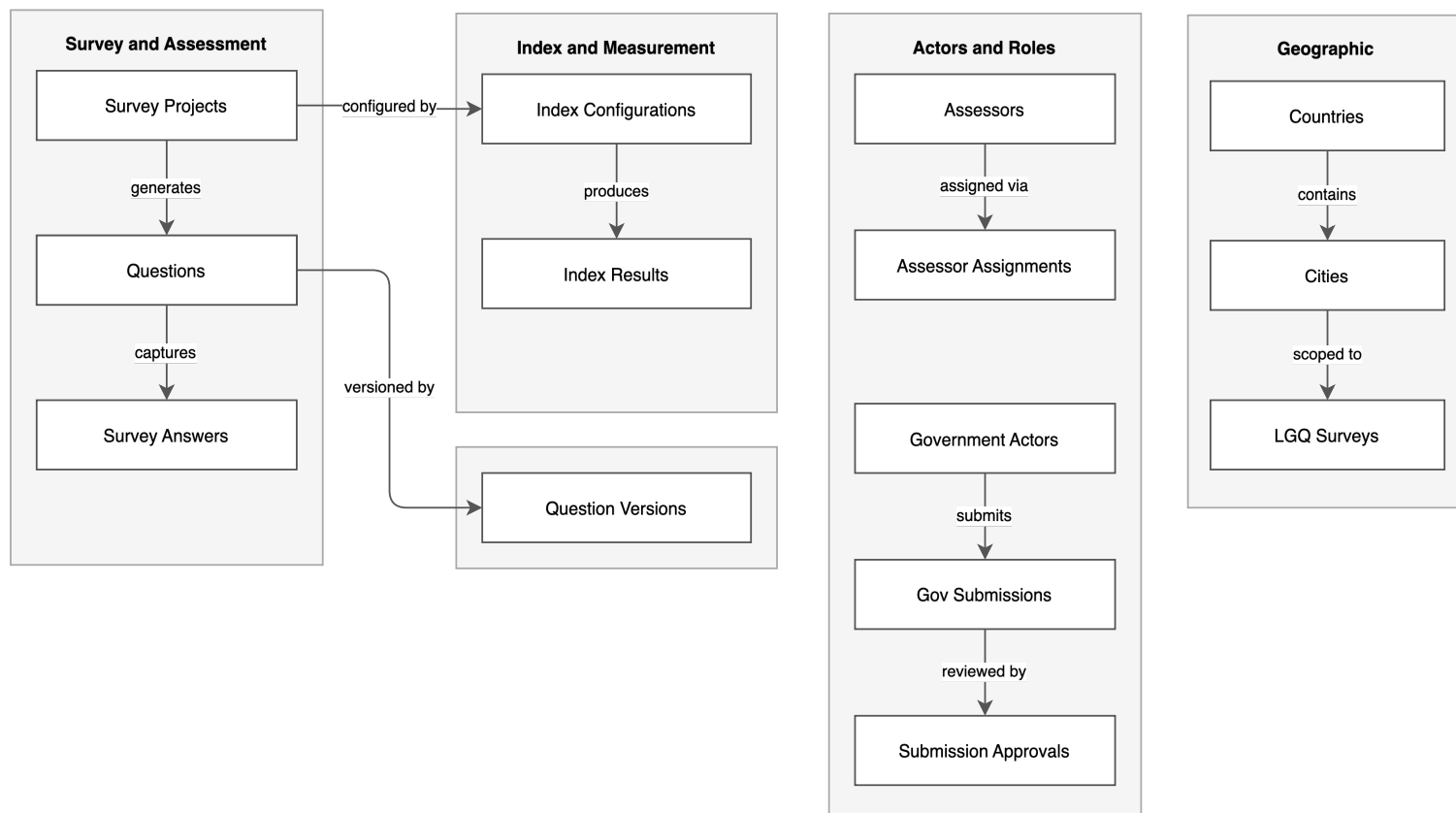
Product	Vendor	ARB Reference	Status	Intended Use in EKAP
Power Platform	Microsoft	2025-112-ARB	ENTERPRISE	Low-code forms, workflows, Power BI dashboards
Azure OpenAI Service	Microsoft	2023-158-ARB	ENTERPRISE	AI/AOSQ in governed scenarios
Microsoft Fabric	Microsoft	2023-393-ARB	ENTERPRISE	Data analytics and BI platform
Qlik Cloud Analytics	Qlik	2024-282-ARB	ENTERPRISE	Alternative analytics platform
Cloudflare CDN	Cloudflare	2024-134-ARB	ENTERPRISE	CDN and perimeter protection
Google Web Analytics	Google	2025-195-ARB	ENTERPRISE	Public portal web analytics
Grafana	Grafana Labs	2023-030-ARB	ENTERPRISE	Operational monitoring
Azure AI Document Intelligence	Microsoft	2025-274-ARB	LIMITED	Document parsing, limited scenarios

5. Database Schema

The EKAP database schema extends the existing EGOVSURVEY relational model (EKBSRVY_ prefix tables) with new entities required for EKAP's expanded functionality. All baseline table definitions are sourced from the Existing DGB Applications and Infrastructure document.



Figure 2: EKAP Database Entity Relationships Overview



5.1 Baseline EGOVSURVEY Schema (EKBSRVY_ Tables)

The following tables constitute the existing EGOVSURVEY database on server NYVM2501 (database: SurveyEgov) and form the migration baseline.

Core Survey and Project Tables

Table	Key Columns	Purpose
EKBSRVY_Projects	SurveyId, Year, Concept, Objective, ExternalDocumentLink, ExternalDocumentTitle, SurveyStartDate, SurveyEndDate, CRDate, CRUser, LMDate, LMUser	Survey cycle definitions and metadata
EKBSRVY_ProjectCountries	ProjectId, CountryId, JuniorResearcher, SeniorResearcher, ResearchStatus, ResearchStatusDate	Assignment of countries to projects with researcher attribution



EKBSRVY_Survey	SurveyId, OnHold, ProjectId, CountryId, ValidFromDate, ValidToDate, StageInDate, JoinAnnDate, LastCertificateDate, CommitDate, FirstNotificationDate, LastNotificationDate, MstNotificationSent, DiscardDate, DiscardBy	Per-country survey instance tracking lifecycle
EKBSRVY_Stages	StageId, Description, OptionNumber, Abbreviation, QuestionPrefix, OrderPosition	OSQ stage definitions
EKBSRVY_Sections	SectionId, ProjectId, Rejected, StageId, OrderPosition	Survey sections aligned to stages

Question and Response Tables

Table	Key Columns	Purpose
EKBSRVY_Questions	QuestionId, Title, Clarification, Type, Threshold, ExternalDocumentLink, ExternalDocumentTitle, CRDate, CRUser, LMDate, LMUser	Master question definitions
EKBSRVY_QuestionOptions	QuestionId, OptionNumber, Description, Abbreviation, QuestionPrefix, Threshold, Text, Value, MinValue, MaxValue, CRDate, Notes	Answer options and scoring thresholds
EKBSRVY_SurveyQuestions	SectionId, QuestionId, OrderPosition, ActiveStatus, ActionDate	Questions assigned to sections within a cycle
EKBSRVY_SurveyAnswer	SurveyId, SectionId, QuestionId, OptionNumber, Notes, ReferenceURLs, CRDate, Discarded, DiscardedBy	Assessor responses with evidence references
EKBSRVY_SurveyQuestionsThemes	SectionId, Questions, ThemeId	Theme classification of questions
EKBSRVY_SurveyQuestionsSubthemes	SectionId, QuestionId, SubthemeId	Sub-theme classification



Classification and Reference Tables

Table	Key Columns	Purpose
EKBSRVY_Countries	CountryId, CountryName, RegionId, SubRegionID, Img_Flag, Img_Map, Income, Link, Population, ISO2Code, PostConflict, DevelopingCountry	Country master data with regional and income classifications
EKBSRVY_Themes	ThemeId, QuestionId, Description, Abbreviation	Question themes (DDVG, ENV, EP, EU, MCSD, OGD, WoG)
EKBSRVY_Subthemes	SubthemeId, ThemeId, Description, Abbreviation	Sub-theme classifications

5.2 EKAP Schema Extensions

New entity groups extend the EGOVSURVEY baseline. Table names use the EKAP_ prefix.

Question Lifecycle and Version History

Table	Key Columns	Purpose
EKAP_QuestionVersions	VersionId, QuestionId (FK), VersionNumber, Title, Clarification, ChangeReason, ValidFrom, ValidTo, CreatedBy, CreatedDate	Full version history per question (History of a Question)
EKAP_QuestionRepopulationRules	RuleId, ProjectId (FK), RepopulationStrategy (BLANK/ALL/VALIDATED), CreatedBy, CreatedDate	Per-cycle repopulation configuration
EKAP_QuestionVersionCycleMappings	MappingId, QuestionId, ProjectId, VersionId, ActiveStatus	Links question versions to specific survey cycles

**OSI/LOSI Toolkit Tables**

Table	Key Columns	Purpose
EKAP_Indicators	IndicatorId, Code, Name, Description, IndexFamily, SubIndex (SP/EPI/IF/CP/TEC), Weight, FormulaReference, VersionNumber, ActiveStatus	Master indicator definitions with index family and weighting
EKAP_QuestionIndicatorMappings	MappingId, QuestionId (FK), IndicatorId (FK), ResponseOptionValue, ContributionWeight, ConditionLogic, VersionNumber, CreatedBy	Links question responses to indicators with scoring logic
EKAP_IndexConfigurations	ConfigId, ProjectId (FK), IndexFamily, ConfigJSON, VersionNumber, ApprovedBy, ApprovalDate, Status (DRAFT/APPROVED)	Versioned OSI/LOSI configuration per survey cycle
EKAP_IndexCalculationRuns	RunId, ConfigId (FK), ProjectId (FK), RunTimestamp, TriggeredBy, Status, OutputSummaryRef	Audit log of each automated index calculation run
EKAP_IndexResults	ResultId, RunId (FK), EntityType (COUNTRY/CITY), EntityId, IndexFamily, SubIndex, Score, InputDataRef, CalculatedAt	Per-entity index results with full input data reference
EKAP_IndexScenarioWorkbench	ScenarioId, BasedOnConfigId, AnalystId, Description, ConfigJSON, RunResultRef, IsSandbox (always TRUE)	Sandbox for testing alternative configurations

Assessor Management Tables

Table	Key Columns	Purpose
EKAP_Assessors	AssessorId, UserId (FK), FirstName, LastName, Email, Languages, Nationality, ExpertiseDomains, EmploymentStatus, RegistrationDate, Status	Assessor master profiles



EKAP_AssessorTraining	TrainingId, AssessorId (FK), CourseCode, CourseName, CompletionDate, Score, PassStatus, CycleApplicable	Training completion records
EKAP_AssessorAssignments	AssignmentId, AssessorId (FK), ProjectId (FK), CountryId (FK), AssignmentType, AssignmentRationale, AssignedBy, AssignedDate, Status	Assignments with documented rationale
EKAP_AssessorPerformance	PerformanceId, AssessorId (FK), ProjectId (FK), CompletionRate, ConsistencyScore, QualityFlags, ReviewerNotes, RecordedDate	Performance scoring per cycle
EKAP_ConflictOfInterest	COIID, AssessorId (FK), CountryId (FK), DeclarationDate, COIType, Resolution, ResolvedBy	Conflict of interest declarations

Government Actor Tables

Table	Key Columns	Purpose
EKAP_GovernmentActors	ActorId, UserId (FK), CountryId (FK), OrganisationName, OfficialTitle, AuthorizationLevel, AuthorizedBy, AuthorizationDate, Status	Authorized government representatives scoped to country or city
EKAP_GovernmentActorSubmissions	SubmissionId, ActorId (FK), ProjectId (FK), InstrumentType (MSQ/LGQ), EntityId, SubmissionData_JSON, SubmissionDate, Status	MSQ and LGQ submission drafts and finals
EKAP_SubmissionApprovals	ApprovalId, SubmissionId (FK), ReviewerId (FK), ReviewerRole, Decision, DecisionNotes, DecisionDate	Approval chain records
EKAP_DiscrepancyCases	CaseId, SubmissionId (FK), AssessmentRef, ValueBefore, ValueAfter, EvidenceRefs, CaseStatus, AssignedTo, ResolutionNotes, ResolvedDate, RecalculationTriggered	Discrepancy cases with before/after values and resolution trail

**External Data and AI Audit Tables**

Table	Key Columns	Purpose
EKAP_ExternalDataSources	SourceId, SourceName, DatasetCode, APIEndpoint, RefreshFrequency, AuthMethod, Status	Registry of external data connectors
EKAP_ExternalDataSnapshots	SnapshotId, SourceId (FK), ProjectId (FK), CaptureDate, DataRef_JSON, IndicatorsMapped, ProviderVersion, ValidationStatus	Per-cycle immutable snapshots for reproducibility
EKAP_ExternalDataIngestionLog	LogId, SnapshotId (FK), RunTimestamp, RecordsIngested, ErrorCount, WarningNotes, TriggeredBy	Audit trail of each ingestion run
EKAP_AOSQInteractions	InteractionId, SessionId, AssessorId (FK), QuestionId (FK), AgentType (PRE_ASSESSMENT/COMPANION/QUA), InputRef, OutputRef, ModelVersion, HumanReviewRequired, ReviewedBy, ReviewDecision, Timestamp	Full audit log of every AOSQ interaction
EKAP_AOSQFeedback	FeedbackId, InteractionId (FK), ReviewerId (FK), FeedbackType, Notes, FeedbackDate	Structured feedback on AOSQ outputs

LOSI Local Government Tables

Table	Key Columns	Purpose
EKAP_Cities	CityId, CountryId (FK), CityName, UrbanisationTier (CITY/TOWN/RURAL), Population, GridDensity, ISO2Code, LOSIScope (LOSI_C/LOSI_T/LOSI_V), Status	City master data with Degree of Urbanisation tier
EKAP_LGQSurveys	LGQSurveyId, ProjectId (FK), CityId (FK), AssessorId (FK), Status, CreatedDate, SubmittedDate	LGQ survey instances per city per cycle
EKAP_LGQAnswers	AnswerId, LGQSurveyId (FK), QuestionId (FK), OptionNumber, Notes, ReferenceURLs, CRDate, Discarded	LGQ responses with evidence references



6. Functional Specifications

Specifications are derived exclusively from the ToR (TOR OP5.1), Feasibility Study, Inception Report, Concept Paper 4.44, and the DGB Infrastructure document.

6.1 Survey Configuration and Questionnaire Builder

Source: ToR functional components; Feasibility Study, Sections 6.1 and 6.3.

Core capabilities:

- Create, configure, and manage survey cycles (projects) with defined start and end dates.
- Build and version questionnaires for OSQ, MSQ, and LGQ instruments within a single platform.
- Assign questions to sections and stages; set order positions, active status, and thresholds.
- Support all six OSQ stages: Stage I Emerging, Stage II Transactional, Stage III Enhanced, Stage IV Connected, Stage V Summarize, Stage VI Linked.
- Configure per-cycle repopulation strategy: blank, prepopulate all previous responses, or prepopulate validated responses only.
- Support duplication of existing projects as the mechanism for new survey cycles, preserving the 20% threshold.

Question lifecycle management:

- Maintain full version history for each question across all cycles (History of a Question).
- Mark questions as active or retired, with retired questions visible for methodological reference.
- Audit trail of all changes to question definitions and mappings.

6.2 MSQ, OSQ, and LGQ Data Collection Workflows

Source: Feasibility Study, Sections 6.1 and 6.5.

- Linked instruments: MSQ, OSQ, and LGQ share common entities for services, portals, and indicators.
- Assessors submit OSQ and LGQ responses through structured data entry interfaces, with required evidence links or attachments.
- Government Actors submit MSQ and LGQ data through the dedicated Government Actor portal.
- Validation rules enforce completeness, consistency, and threshold compliance before submission.
- Discrepancy review workflow between Government Actor self-assessments, assessor findings, and automated indices, including escalation paths and documented resolutions.



- Discrepancy cases logged with before-and-after values, evidence references, reviewer details, and a recalculation trigger once resolved.

6.3 Assessor Management and Lifecycle

Assessor profiles:

- Languages, expertise domains, location, employment status, previous participation, and training completion.
- Conflict-of-interest declarations per country and per cycle.

Eligibility and assignment:

- Eligibility rules combining minimum training completion, performance thresholds, and conflict-of-interest checks.
- Automated assignment based on language, region, expertise, and workload, with manual override and logged rationale.
- Ability to restrict or suspend assessors based on persistent quality issues.

Training and performance:

- Training completion tracking per assessor per survey cycle.
- Performance dashboards showing completion rates, consistency scores, and quality flags.

6.4 Government Actor Portal

Portal capabilities:

- Log in and access only the country or city for which they are authorized.
- Submit and update MSQ and LGQ data through structured forms.
- Draft updates for country and city knowledge base pages in EGovKB.
- Save drafts, submit, and track submission status and approval outcomes.

Approval chain:

- All submissions route through a structured approval chain to UN analysts and senior approvers.
- Complete audit trail of submission, review, and approval or rejection with documented rationale.

6.5 OSI/LOSI Toolkit and Index Governance

Source: Feasibility Study, Section 7; Concept Paper 4.44.

- OSI formula: $OSI = 0.45 \times SP + 0.35 \times EPI + 0.10 \times IF + 0.05 \times CP + 0.05 \times TEC$.
- LOSI tiers: LOSI-C (cities above 50,000), LOSI-T (towns above 5,000), LOSI-V (rural areas below 300/km²).



- HCI incorporates EGL from 2024 (UNESCO and UNDP data). TII data sourced from ITU.
- Automated calculation of scores for services, portals, cities, and countries after data approval.
- Measurement configuration per cycle specifying which MSQ, OSQ, LGQ, and external data points contribute to each index.
- Sandbox scenario workbench for testing alternative mappings or weights without affecting official results.
- Contextual links from MSQ, OSQ, and LGQ workspaces to relevant National and Local E-Government Toolkit modules.

6.6 Automated Index Measurement and Data Approval

- Automated generation of OSI, LOSI, EGDI, and related indices from validated MSQ, OSQ, LGQ, and external data.
- Data approval workflows where DGB validates input data and finalizes configurations before triggering calculation.
- Every calculation run logged with configuration version, triggering user, timestamp, and output reference.
- Comprehensive audit logs of index configuration changes, approvals, and calculation runs.

6.7 AI Services and AOSQ

AOSQ (Agent-assisted Online Services Questionnaire layer) provides three governed agent functions:

- Pre-assessment agent: pre-screens public portal evidence and prepares draft observations for assessor review.
- Companion agent: assists assessors during OSQ completion with contextual guidance and consistency checks.
- Quality assurance agent: flags inconsistent answers, missing evidence, or potential anomalies.

Governance requirements:

- Human-in-the-loop decision-making is mandatory; AI supports but does not replace expert judgement.
- All AI interactions logged in EKAP_AOSQInteractions for full traceability and audit.
- Azure OpenAI Service (ARB 2023-158, ENTERPRISE). No AI involvement in official index scoring.

6.8 Knowledge Base and EGovKB Integration

- EGovKB continues as the primary outward-facing knowledge portal, consuming EKAP data via APIs.
- EKAP publishes country and city profiles, EGDI/LOSI/OSI/TII/HCI results, and toolkit updates to EGovKB.



- Dashboards for EGDI, OSI, LOSI, HCI, and TII with filters for region, income group, CSS groups (LDC, LLDC, SIDS), cycle, country, and city.
- Configurable exports using standard templates for EGDI and LOSI publications.

6.9 Reporting and Analytics

- Analytical dashboards for EGDI, OSI, LOSI, HCI, and TII with full filter capabilities.
- Integration with Power BI (ARB 2025-112) or Microsoft Fabric (ARB 2023-393) for advanced analytics.
- Grafana (ARB 2023-030) for operational system monitoring metrics.
- Configurable data exports in formats required for EGDI and LOSI publications.

6.10 User Management and Role-Based Access Control

Role	Permissions
UNV / Junior Reviewer	Complete OSQ and LGQ assessments; view assigned country data; submit responses with evidence.
Senior Reviewer	Review and validate assessor submissions; flag discrepancies; access quality dashboards.
Manager / DGB Analyst	Configure survey cycles; manage assessor assignments; approve index configurations; access all dashboards.
Administrator	Full system administration; user management; configuration management; RBAC assignment.
Government Actor	Access assigned country or city only; submit MSQ and LGQ updates; track submission status.
Public User	Read-only access to published EGovKB profiles and EGDI/LOSI dashboards.

- Authentication via UNHQ Active Directory (NOT Azure AD); VPN required for administrative access.
- Citrix NetScaler gateway mediates external web access for assessors and Government Actors.

6.11 External Data Integration

- Configurable connectors for World Bank, ITU, UNESCO, UNDP, and other trusted international organizations.
- Per-cycle immutable snapshots ensuring any EGDI or LOSI result can be reproduced using the exact data used at calculation time.
- Administrative dashboards for monitoring data refreshes, ingestion errors, and manual overrides.



6.12 Document Management

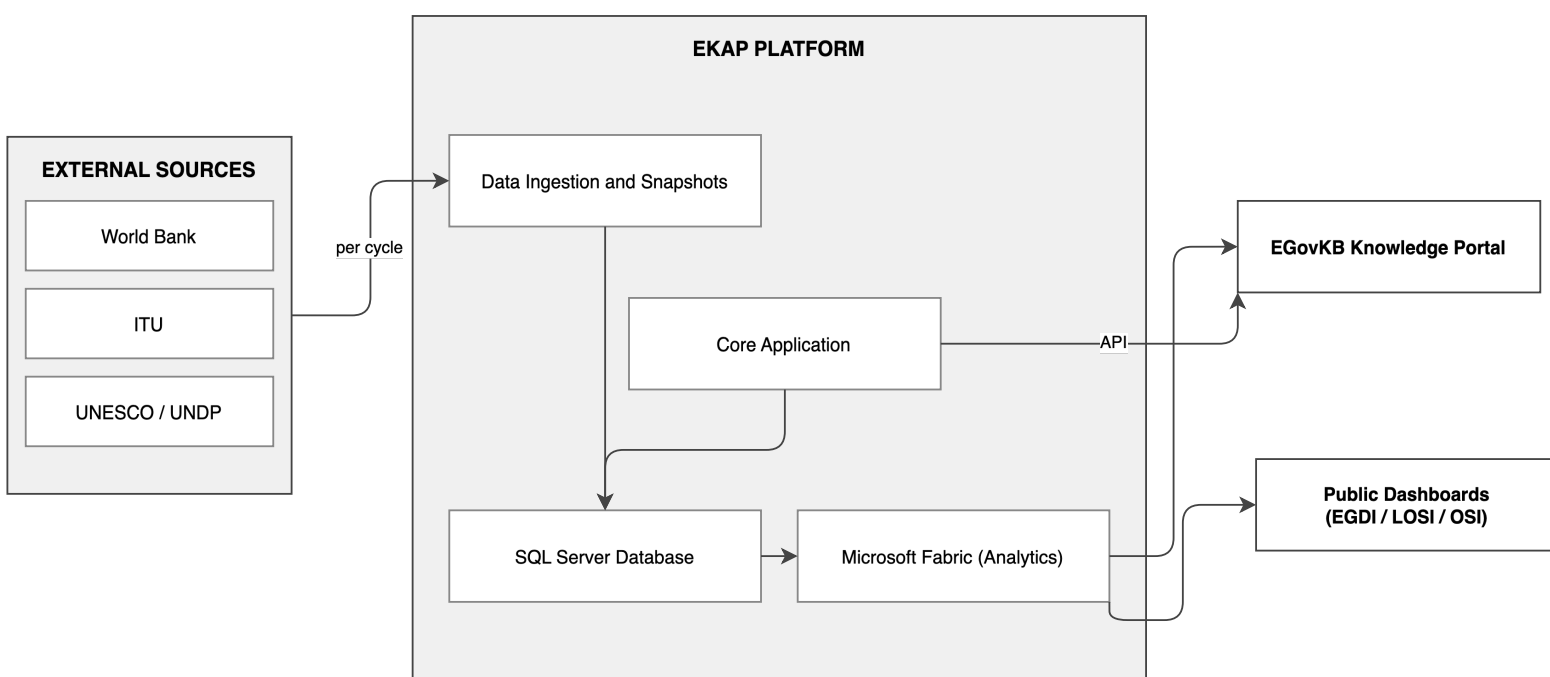
- Structured repository for survey methodology documents, assessor guidance, country evidence files, and reports.
- Version control for document uploads with metadata capture.
- Evidence attachments linked to specific survey answers and OSQ responses.
- Access control aligned with RBAC roles; Government Actor documents scoped to assigned country.

7. Integration Architecture

7.1 Inbound External Data Integration

Source	Data Provided	Method	Frequency
World Bank	Country development indicators: income, GDP, population	REST API	Per survey cycle
ITU	Telecommunication Infrastructure Index (TII) data	REST API / file ingestion	Per survey cycle
UNESCO	Human Capital Index components; education data; Expected Gross Literacy (EGL) from 2024	REST API / file ingestion	Per survey cycle
UNDP	Human Development Index components contributing to HCI	REST API / file ingestion	Per survey cycle

Figure 3: EKAP Integration Architecture





7.2 Outbound Integration

- EKAP exposes a secure API layer consumed by EGovKB for country profiles, city profiles, and indicator data.
- Published EGD, OSI, LOSI, TII, and HCI results flow from EKAP to EGovKB after data approval.
- Toolkit indicator definitions and formula references linked from EKAP to EGovKB toolkit pages.

7.3 Security and Network Integration

- Citrix NetScaler gateway for external-facing application access.
- UNHQ VPN required for all administrative and infrastructure management access.
- Cloudflare CDN (ARB 2024-134) for public-facing portal performance and perimeter protection.
- Monitoring and logging aligned with OICT operational standards.

8. Security and UN OICT Compliance

8.1 Security Design Principles

- Privacy-by-design: data minimization, anonymization where possible, and strict access controls.
- Secure coding practices aligned with OICT standards.
- Regular security testing throughout development and before production deployment.
- All external-facing components protected through Citrix NetScaler gateway and Cloudflare CDN.

8.2 Identity, Access, and Data Protection

- Authentication via UNHQ Active Directory; no implicit access granted.
- Government Actor access strictly scoped to authorized country or city.
- Personal information governed by UN data protection policies.
- Data residency requirements confirmed with OICT for all cloud-hosted components.
- Contractual safeguards for any external AI or SaaS services handling UN data.

8.3 AI Governance

- All AI services operate within OICT-approved boundaries (Azure OpenAI Service, ARB 2023-158).
- Human oversight mandatory for all AI-generated outputs.
- AI use restricted to approved scenarios; no AI involvement in official index scoring.

8.4 Audit and Monitoring



- Comprehensive audit logs: configuration changes, data approvals, discrepancy decisions, calculation runs, and access events.
- Grafana (ARB 2023-030) for operational system monitoring, health checks, and alerting.

9. Non-Functional Requirements

9.1 Performance and Scalability

- Concurrent access during peak survey periods without degradation.
- Scalable cloud architecture using autoscaling on Azure or AWS.
- Load testing required before production launch.

9.2 Availability and Reliability

- Production availability aligned with UN operational standards.
- Robust backup and rollback procedures throughout migration and operation.
- Disaster recovery plan with defined RTO and RPO, subject to OICT standards.

9.3 Multilingual and Accessibility

- Multilingual support aligned with UN language requirements.
- Accessibility compliance with applicable UN web accessibility standards.

9.4 Maintainability and Extensibility

- Modular architecture enabling independent updates to each module.
- API-first design enabling future integrations without core redesign.
- Version control for all configuration, schema changes, and index definitions.

10. Data Migration Approach

10.1 Migration Scope

- Migrate all EGOVSURVEY EKBSRVY_ tables from NYVM2501 (database: SurveyEgov) into the EKAP data model.
- Preserve all historical OSQ, MSQ, and LGQ responses across prior survey cycles.
- Preserve historical EGDI, OSI, LOSI, TII, and HCI index values for longitudinal analysis.
- Migrate EGovKB content structures to enable API-based consumption.

10.2 Migration Steps

- Map EGOVSURVEY and EGovKB data to the EKAP schema using the EKBSRVY_ table inventory.
- Migrate question history including all versions, index mappings, and cycle usage records.



- Execute post-migration validation: record counts, key integrity, sampling verification, and index spot checks.
- Plan cut-over between survey cycles to avoid gaps in reporting continuity.

10.3 Risk Management During Migration

- Run dual systems (legacy EGOVSURVEY and EKAP) in parallel during a defined transition period.
- Maintain robust database backup and rollback procedures throughout migration.
- Stage all migration activities in testing and staging environments before production cut-over.

11. Implementation Roadmap

11.1 Phased Implementation Beyond the Consultancy

Phase 1 - Foundation:

- Implement the integrated EKAP data model and minimal survey engine.
- Consolidate EGOVSURVEY and EGovKB data into EKAP.
- Implement basic dashboards and exports.

Phase 2 - Toolkit and Roles:

- Implement OSI/LOSI toolkit and index governance modules.
- Implement assessor lifecycle management and Government Actor portal.
- Strengthen EGOVSURVEY-EGovKB integration through EKAP APIs.

Phase 3 - Optimization:

- Integrate AI and AOSQ features for limited, well-governed scenarios.
- Expand BI and analytics capabilities.
- Enhance learning and toolkit integration.

11.2 Alignment with ToR Required Functional Components

ToR Required Component	Section in This Document
Interactive Survey Builder	Section 6.1 - Survey Configuration and Questionnaire Builder
Multi-Tiered Evaluation System	Section 6.2 - MSQ, OSQ, LGQ Workflows; Section 6.10 - RBAC
Survey Data Analysis Dashboard	Section 6.9 - Reporting and Analytics
Document Management	Section 6.12 - Document Management
User Management and RBAC	Section 6.10 - User Management and RBAC
API Integration Layer	Section 7 - Integration Architecture



Knowledge Base Module	Section 6.8 - Knowledge Base and EGovKB Integration
AI-driven Modules	Section 6.7 - AI Services and AOSQ
Secure Scalable Infrastructure	Section 4.4 - Hosting; Section 8 - Security and Compliance

12. Design Adjustments from Technical Review

12.1 Government Actor as a Formal Named Role

Government Actor is a formal, explicitly named role in EKAP's RBAC model, not subsumed within a generic external stakeholder category. All access controls, workflow routing, portal interfaces, and audit logging treat Government Actors as a distinct entity type scoped to an authorized country or city.

12.2 LGQ Standardization

The Local Government Questionnaire (LGQ) is a standardized instrument within EKAP, parallel in structure to OSQ and MSQ. LGQ surveys are scoped using the Degree of Urbanisation methodology. EKAP_LGQSurveys and EKAP_Cities enforce consistent tier classification (LOSI-C, LOSI-T, LOSI-V) across all LGQ instances.

12.3 AOSQ Terminology

AOSQ is formally defined as Agent-assisted Online Services Questionnaire layer, comprising three agent functions: pre-assessment, companion, and quality assurance. This aligns with Concept Paper 4.44 and is used consistently throughout Section 6.7.

12.4 Admin and Configuration Layer

A dedicated administrative and configuration layer is specified within the Application Layer covering: survey cycle configuration, OSI/LOSI toolkit version management, index configuration approval, assessor management administration, and system-wide RBAC assignment.

12.5 External Data Model Scope

The external data integration model accommodates the full range of trusted external sources referenced in the Inception Report 2.4: World Bank, ITU, UNESCO, UNDP, and other international organizations as designated by DGB.

12.6 Infrastructure-Neutral Architecture

The target architecture supports both Azure and AWS hosting, ensuring compatibility with any future OICT hosting decision. No single-vendor lock-in is introduced.

12.7 LOSI City Publication



LOSI results for individual cities are included in the EKAP publication workflow. City naming follows the existing convention: CountryCode - CityName (example: AE - Dubai), as documented in the DGB Infrastructure document.

13. Conclusion and Next Steps

This Technical Design Package establishes the architecture plan, database schema, and functional specifications for EKAP as required under Deliverable 3 of TOR OP5.1 EKAP. All specifications are grounded in the Inception Report, the Feasibility Study, Concept Paper 4.44, and the existing DGB infrastructure, and are aligned with UN OICT standards and the approved product registry.

- The Visual Mock-Up of the Platform will translate the functional specifications in Section 6 into visual interface designs covering the assessor interfaces, Government Actor portal, UN staff dashboards, and public portal outlined in Section 4.3.
- The Prototype Development and Testing deliverable will implement a working prototype based on the hybrid architecture defined in Section 4, the database schema in Section 5, and the functional specifications in Section 6.
- The Implementation Terms of Reference will define the procurement scope and technical requirements for full EKAP implementation, drawing on the RBAC model in Section 6.10, the integration architecture in Section 7, and the phased approach outlined in Section 11.2.
- Data migration from EGOVSURVEY (NYVM2501, database SurveyEgov) to the EKAP schema will follow the approach defined in Section 10.
- UN DESA and UN OICT are invited to review this Technical Design Package and provide formal feedback before prototype development commences.

Annex A: OSQ Stage Definitions

Source: Existing DGB Applications and Infrastructure document (EKBSRVY_Stages table).

Stage	Description	Abbreviation	Definition
Stage I	Emerging	I	Basic web presence; foundational government portal features.
Stage II	Transactional	II	Online transactional services; citizen-facing digital interactions.
Stage III	Enhanced	III	Enhanced services with multi-channel delivery and digital integration.
Stage IV	Connected	IV	Connected government; cross-agency integration and interoperability.
Stage V	Summarize	V	Summary and synthesis stage for comprehensive service evaluation.
Stage VI	Linked	VI	Fully linked and integrated digital government ecosystem.



Annex B: OSI Formula and Sub-Index Weights

Source: Concept Paper 4.44.

$$\text{OSI} = 0.45 \times \text{SP} + 0.35 \times \text{EPI} + 0.10 \times \text{IF} + 0.05 \times \text{CP} + 0.05 \times \text{TEC}$$

Sub-index	Name	Weight	Description
SP	Service Provision	45%	Measures breadth and depth of online government services available to citizens.
EPI	E-Participation Index	35%	Measures quality and scope of digital citizen engagement and participatory mechanisms.
IF	Infrastructure	10%	Measures the technological infrastructure supporting online service delivery.
CP	Content and Policy	5%	Measures accessibility and completeness of government content and policy frameworks.
TEC	Technology	5%	Measures use of emerging and enabling technologies in government service delivery.